Module 5

Configuring Routing and Advanced Switching

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Learning Objectives

- Compare and contrast routing concepts.
- Compare and contrast dynamic routing concepts.
- Install and troubleshoot routers.
- Explain tiered switching architecture.
- Explain virtual LANs.

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Lesson 5.1

Routing Technologies

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Routing Tables and Path Selection

The following main parameters define a routing entry:

- Protocol
- Destination
- Interface
- Gateway/next hop

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Static and Default Routes

Routing table entries

- Directly connected routes
- Remote routes
- Host routes
- Default route

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Routing Table Example

Router B Routing Table		
Network	Interface	Source
10.0.1.0/24	G0	Static
10.0.2.0/24	G0	Connected
10.0.3.0/24	G1	Connected
10.0.4.0/24	G1	Static

Router A Routing Table		
Network	Interface	Source
10.0.1.0/24	G0	Connected
10.0.2.0/24	G1	Connected
10.0.3.0/24	G1	Static
10.0.4.0/24	G1	Static

Router C Routing Table		
Network	Interface	Source
0.0.0.0/0	G0	Static
10.0.3.0/24	G0	Connected
10.0.4.0/24	G1	Connected

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Packet Forwarding

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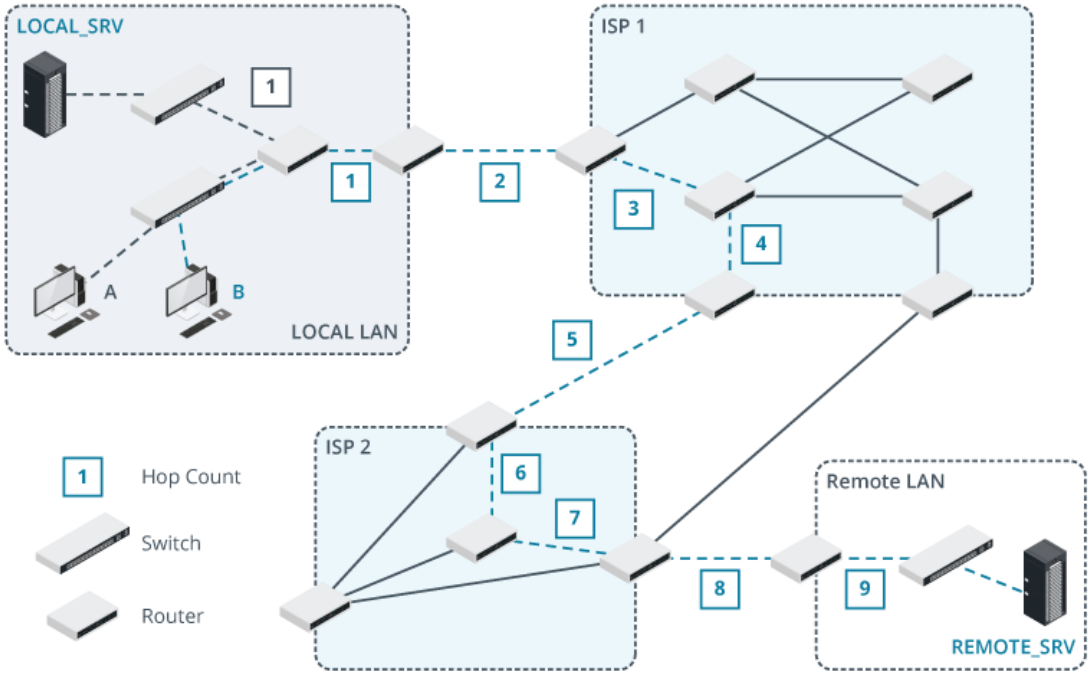
Hop Count



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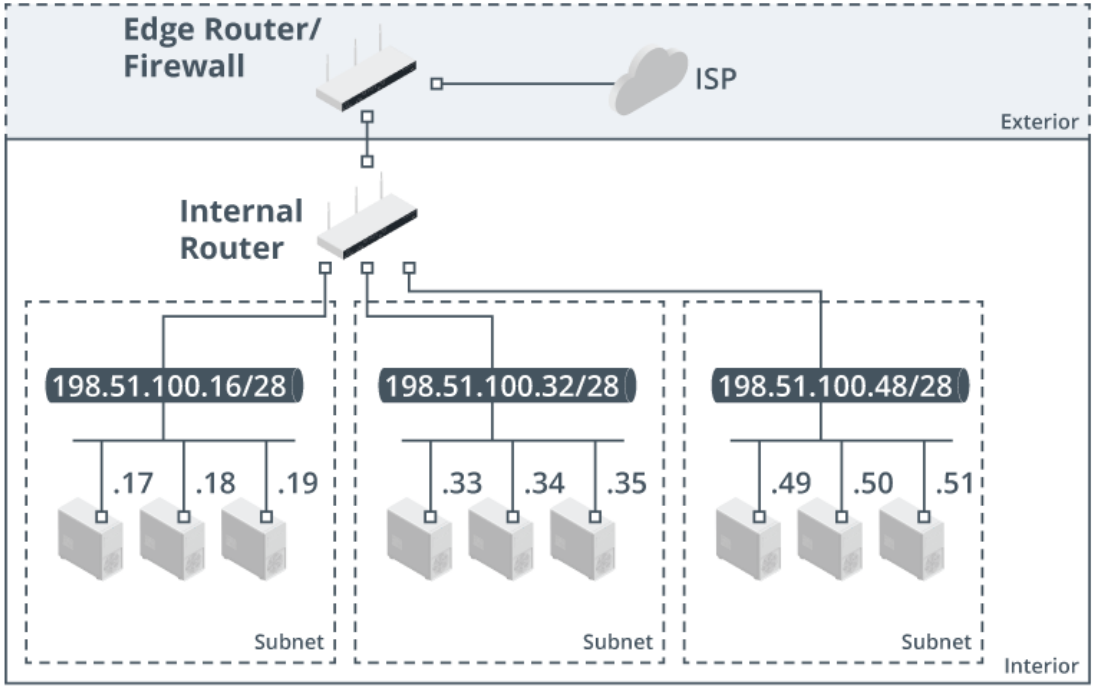
Router Configuration

Router placement:

- Same subnet or IP network must not be separated by a router
- Different subnets or IP networks must be separated by a router



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Routing Tools

show route	show arp	route	traceroute
• Displays routing table	• Lists current ARP table entries	• Shows IP routing table of a host	• Tracks a packet's path to destination

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Activity: Trivia

What is the function of a routing table?

What are directly connected routes?

What is traceroute used for?

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Dynamic Routing Technologies

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Static vs. Dynamic Routing

Feature	Static Routing	Dynamic Routing
Configuration	Manually configured	Automatically adjusts to network changes
Flexibility	Inflexible - Updates require manual intervention	Flexible - Adapts in real-time
Control	Complete control over routing paths	Less direct control over the routes that data takes
Use Case	Ideal for small, stable networks where routes do not change often	Ideal for larger, more complex networks with frequent changes

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Dynamic Routing Protocols

Definition

- Dynamic routing protocols are algorithms that automatically update route information and adjust the paths between network nodes by distributing network topology information.

Advantages

- Scalability, adaptivity to network changes, and reduced network administration overhead

Considerations

- Requires more processing power and memory
- Proper configuration is critical for security and efficiency

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RIP Protocol

Definition

- A legacy dynamic routing protocol that finds the best path between the source and destination networks.

Features

- Uses hop count as the metric for path selection
- 15 maximum allowed hops

Considerations

- Not ideal for large networks because of the hop count limit
- Slow to converge in response to network changes

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Enhanced IGRP (EIGRP)

Definition

- An advanced distance-vector protocol that is used on a computer network for automating routing decisions and configurations

Features

- Uses metrics such as bandwidth, delay, load, and reliability for path selection
- Supports both IPv4 and IPv6 without needing separate configurations

Considerations

- More complex to configure
- Cisco Systems proprietary protocol

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Open Shortest Path First (OSPF)

Definition

- A dynamic link-state protocol that efficiently exchanges routing information within an autonomous system using the Shortest Path First algorithm.

Features

- Computes the shortest path first
- Supports complex network topologies

Considerations

- Can be complex planning and configuration
- Requires more bandwidth than distance-vector protocols

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Border Gateway Protocol (BGP)

Definition

- Gateway protocol that enables the Internet to exchange routing information between autonomous systems.

Features

- Scalable to the Internet’s size, handling thousands of routes
- Uses path vector protocol for establishing routing decisions
- Supports CIDR, allowing for efficient IP address management and route aggregation.

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Activity: Fill in the Blank

1. _____ are algorithms that automatically update route information and adjust the paths between network nodes by distributing network topology information.

2. _____ is a protocol that enables the Internet to exchange routing information between autonomous systems.

3. _____ is a legacy dynamic routing protocol that finds the best path between the source and destination networks.

4. _____ is a dynamic link-state protocol that efficiently exchanges routing information within an autonomous system using the Shortest Path First algorithm.

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Network Address Translation

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Network Address Translation (NAT)

What it does

- Modifies the network address information in packet headers while in transit
- Enables multiple devices on a local network to share a single public IP address

Why it is needed

- Increases network security by hiding internal IP addresses from external networks
- Conserves public IP addresses

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NAT Types

Dynamic

Maps internal addresses to a pool of external addresses dynamically

Allows a larger number of devices to share the same external IP address

Static

Translates one internal IP address to one external IP address.

Allows inbound connections initiated from outside the network.

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Edge Routers

An edge router

- Is located at the boundary of a network that connects to external networks (e.g., the Internet).
- Manages the flow of data between the internal and external networks.
- Routes data to its destination, performs NAT and applies security measures.

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Types of Edge Routers

Customer Edge (CE)

Located at the edge of a customer's network

Connects customer's internal network to provider's network, acting as a security barrier for the customer's network

Provider's Edge (PE)

Located at the edge of a provider's network

Focuses on maintaining the integrity and confidentiality of customer data as it travels across the provider's network

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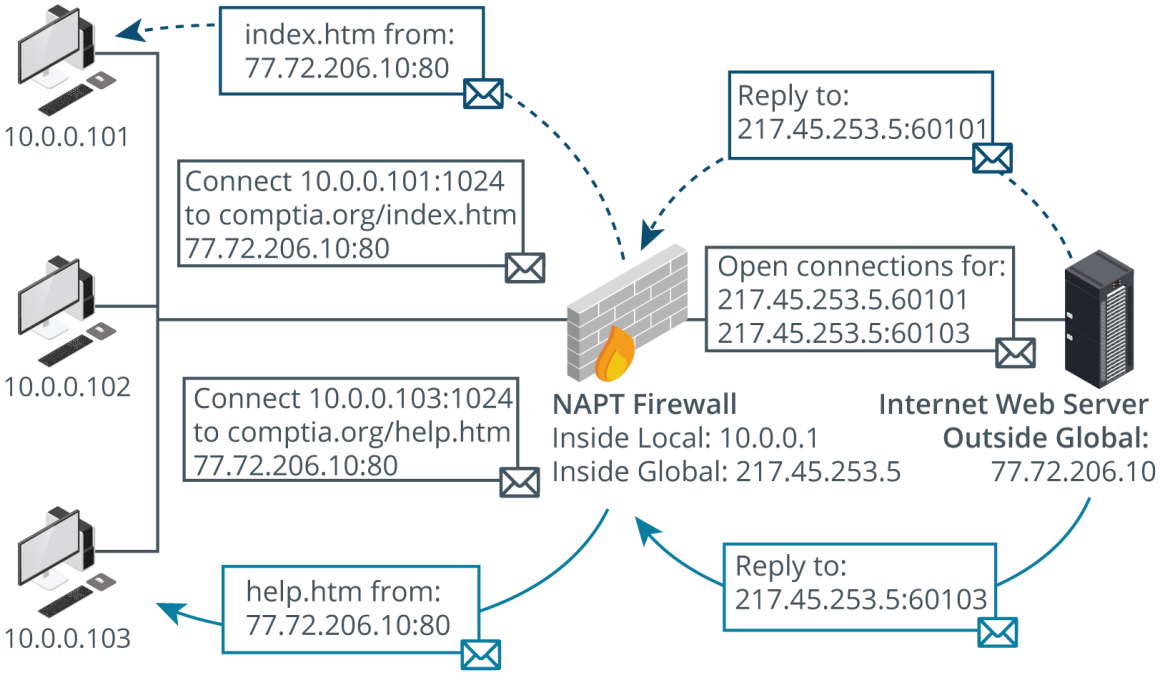
Port Address Translation



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Activity: Multiple Choice

Conserves public IP addresses

Maps internal addresses to a pool of external addresses

Maintains the confidentiality of a customer's data as it travels across the provider's network

NAT

PAT

Dynamic NAT

Static NAT

CE

PE

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Firewalls

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Firewall Types

Hardware

- Standalone appliances
- Integrated within routers
- Next-Generation (NGFW)

Software

- Operating system-based
- Third-party software-based

Cloud-based

- Firewall as a service (FAAS)

Unified Threat Management (UTM)

- All-in-one security appliances

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Stateless vs. Stateful

Stateless inspection (Packet filtering)

- Filters based on IP addresses, protocol, and port numbers
- Acts at the network layer
- Suitable for smaller networks or less complex security requirements

Stateful inspection (Circuit Level Gateway)

- Inspects packets and tracks the state of active connections
- Acts at the session layer
- Ideal for corporate networks where it is critical to protect sensitive data

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Activity: Window Shopping

Barricuda.com

Checkpoint.com

Fortinet.com

Research 3 different types of routers noting the different types, volume of traffic supported, where it should be placed on a network, and any other notable features.

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Enterprise Networking Topologies

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Activity: Think About It

What are the three basic network topologies?

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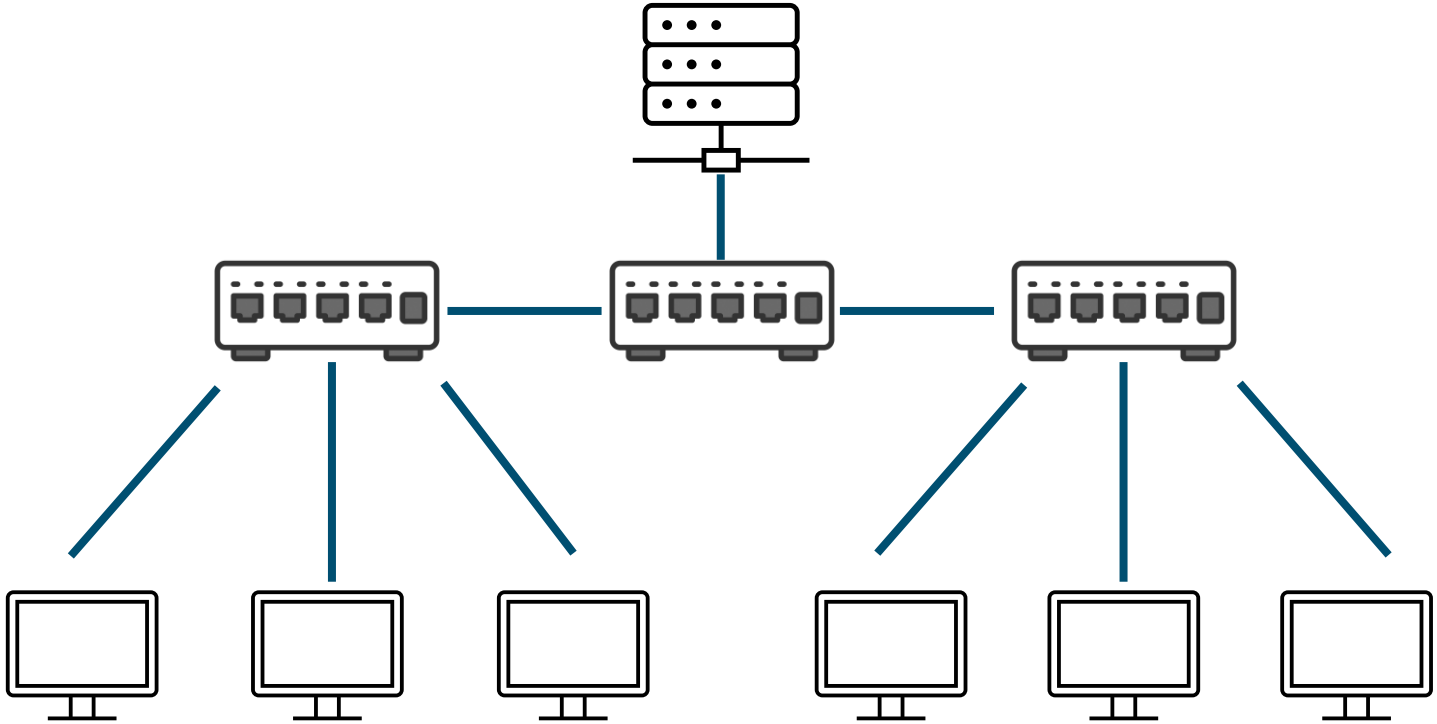
Tree Topology



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Hybrid Topologies

Hybrid Topologies

- Used when a basic topology isn’t enough
- Use a mixture of the basic topologies
- Used to implement redundancy and fault tolerance

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Activity: Quick Search

- Research these hybrid topologies and note the following:
- What do they look like?
- How and when are they used?

Hierarchical star

Hierarchical star-mesh

Star of stars

3-Tiered network hierarchy

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Virtual LANS

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Virtual LANs and Subnets

Creates separate networks within a single physical network infrastructure

Segmentation

- Divides a network into small, isolated segments
- Doesn’t require separate hardware

Flexibility

- Users and devices are grouped by function, department or team

Simplified Administration

- Network changes or moves can be easily managed without altering physical setup

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VLAN IDs and Membership

VLAN IDs

- Each VLAN is assigned a unique ID
- Each device is a member of a VLAN
- Devices use VLAN IDs to manage traffic
- Inter-VLAN routing is needed for devices to communicate between VLANs

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Tagged and Untagged Ports

- Port tagging: method of marking data packets with a VLAN ID
- Tagged ports: configured to receive and understand packets that have a VLAN identifier
- Untagged ports: set to receive data without a VLAN identifier

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VLAN Types

Default VLAN	Native VLAN	Voice VLAN	Management VLAN
• VLAN with ID1 • Should remain unused • Change unused ports from ID1	• Receives all untagged frames • Separate Native and Default VLANs • Match Native VLAN IDs on switches	• Handles VoIP system traffic • Ensures voice communication quality • Prioritizes voice traffic	• Manages network devices and services • Isolate from user data traffic • Separate from Default VLAN

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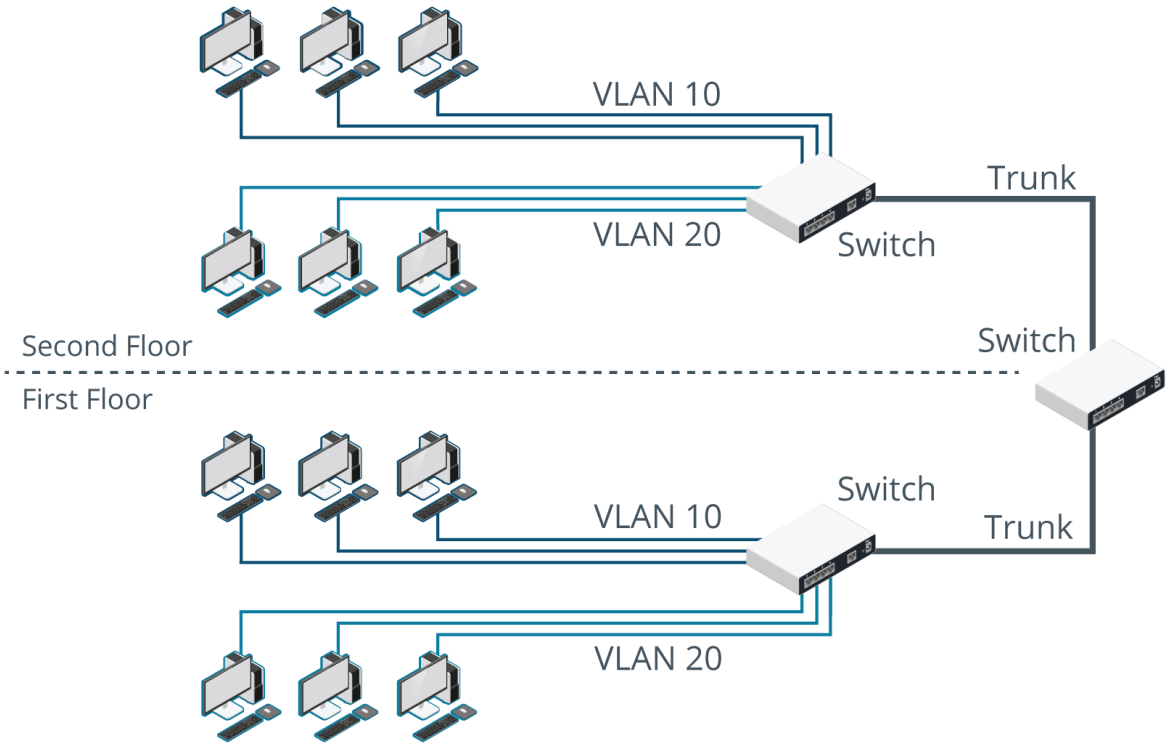
Trunking and IEEE



Trunking bundles physical links into one logical link

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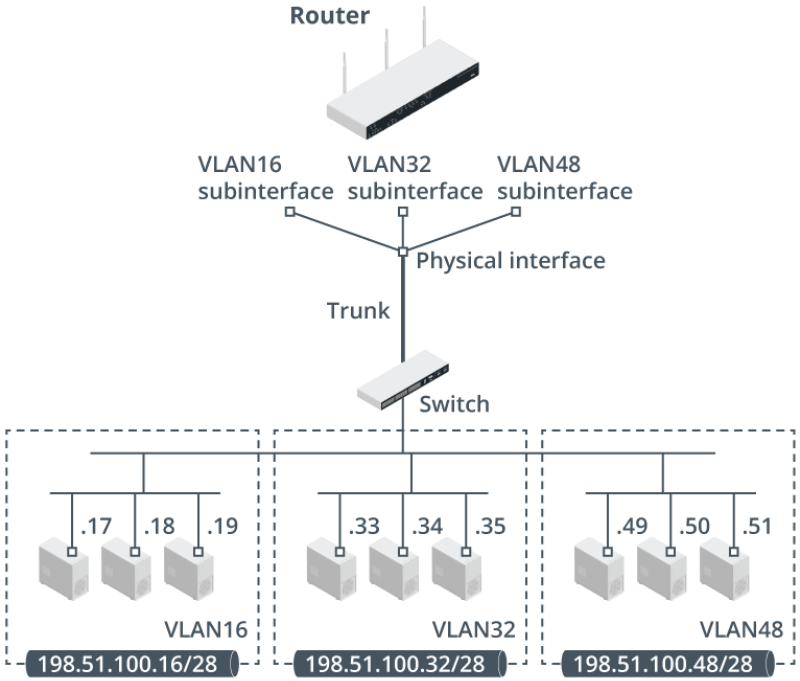
VLAN Routing



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Activity: Two Truths and a Lie

Each device in a VLAN is assigned a unique VLAN ID.

Untagged ports are set to receive data without a VLAN identifier.

A Native VLAN receives all untagged frames.

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Routing and VLAN Troubleshooting

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Routing Table Issues

Suspect a routing issue if you ping a host's default gateway but not some or all hosts on remote network.

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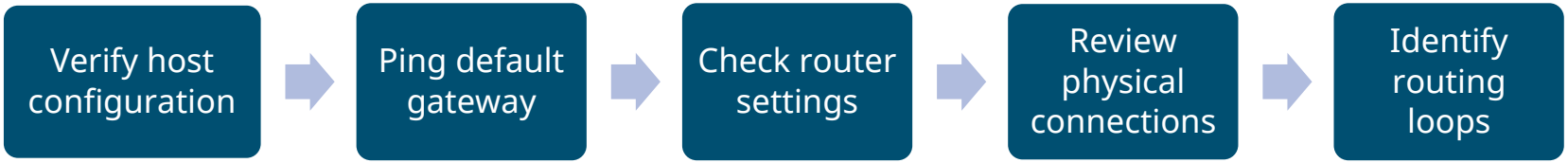
Default Route and Routing Loop Issues

Misconfigured default routes or distributing default routes to other routers can lead to routing loops.



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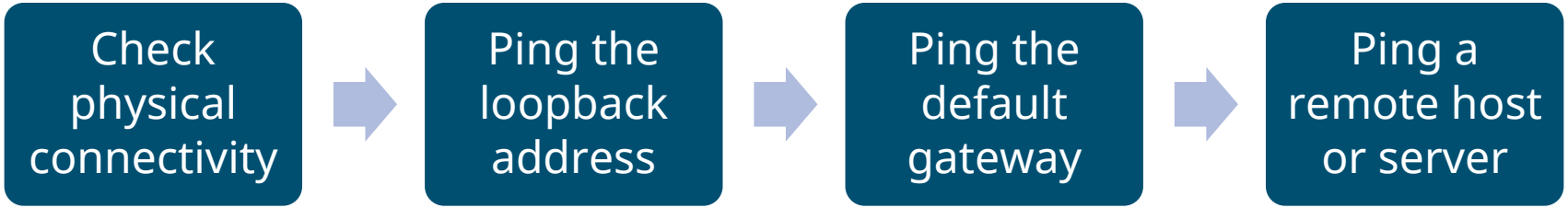
VLAN Assignment Issues

Devices on a VLAN need a matching IP configuration (address, subnet mask, gateway, DNS).



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Summary

- **Routing Tables**: Routers' internal maps guide data packets to their destinations
- **Dynamic Routing**: Protocols automate routing information exchange between routers
- **NAT**: Translates between private and public IP addresses for internet access
- **VLANs**: Create logical network segments within a physical network

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